AI-NATIVE GITHUB

10 OUT OF 10

The Complete Architecture Guide

What a perfect-score GitHub setup looks like for teams running AI agents

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THE REALITY CHECK

- ✗ You don't need a DevOps team
- ✗ You don't need enterprise GitHub
- ✗ You don't need 100 repos

WHAT YOU NEED

- Clear branch model
- AGENTS.md in every repo
- Branch protection rules
- Conventional commits enforced

14 DIMENSIONS OF EXCELLENCE

- Org metadata
- Repo descriptions
- Branch model
- Branch protection
- Architecture scaffold
- Version control
- CHANGELOGs
- AGENTS.md
- PR templates
- Issue templates
- Commit conventions
- Release protocol
- Documentation
- Teams & permissions

1 STEP 1: ORGANIZATION SETUP

Teams Structure

- engineering — Write on all repos
- agents — Write on assigned repos
- reviewers — Approve merges
- admin — Org owner

Org Metadata

- Name & description
- Contact email
- Website URL
- Profile README
- Default branch = dev

2 STEP 2: REPOSITORY ARCHITECTURE

Full Repository Scaffold:

```
product/  (shippable code: api, services, workers, ui)
agents/   (OpenClaw and Claude Code configs)
prompts/  (version-controlled prompts)
specs/    (task files, feature briefs)
tests/    (all tests)
infra/    (LOCKED — deployment, CI)
docs/     (architecture, runbooks, ADRs)
.github/  (CODEOWNERS, templates, workflows)
```

PRO TIP: Don't scaffold everything. Skills repos, config repos, and templates use lighter structures.

3 STEP 3: BRANCH MODEL

Protected Branch Hierarchy:

```
main (protected)  ← tagged releases only
  ↑
dev (protected, default)  ← integration, always deployable
  ↑
human/[name]/TASK-XXX-slug  ← human feature branches
agent/TASK-XXX-slug         ← AI agent work
sandbox/[experiment]        ← exploration, never merges
hotfix/[slug]               ← production emergencies
```

CRITICAL RULES:
OpenClaw never touches human/* branches. Humans never commit to agent/* branches. 48-hour branch TTL.

4 STEP 4: AGENTS.MD — THE AI STANDARD

Key Sections Every AGENTS.md Must Have:

- Scope — which files agents can modify
- Branches — naming conventions and restrictions
- Commits — format and frequency rules
- Session Rules — duration, task limits
- Self-Fix Protocol — how to handle CI failures
- Review Requirements — risk tiers and approval paths

20,000+ repos have adopted this standard. It's the emerging GitHub norm for AI-native teams.

5 STEP 5: BRANCH PROTECTION & ENFORCEMENT

4 Critical Hooks:

- Conventional Commits — enforce via git hooks or CI
- Scope Checking — agent PRs stay within declared files
- Auto-Labeling — agent-generated, high-risk, etc.
- Risk Scoring — diff size, file locations, scope compliance

Branch Protection Rules

```
main: 1 approval, CI required, no admins override
dev: 1 approval, CI required, admin can emergency merge
```

6 STEP 6: RELEASE PROTOCOL & SEMVER

8-Step Release Sequence:

- All target PRs merged to dev
- dev is stable, CI passing, tested
- Create release PR: dev → main
- Owner reviews and approves
- Merge to main (merge commit, NOT squash)
- Tag: git tag -a vX.Y.Z -m 'description'
- Push tag: git push origin vX.Y.Z
- Create GitHub Release with notes

Golden Rule: One product = one repo. Versions are tags, not repos.

7 STEP 7: CI/CD FOR AI AGENTS

Agent-Specific CI Checks:

- Scope Checking — verify agent didn't touch restricted files
- Self-Fix Loop — on CI failure, feed error back to agent
- Auto-Labeling — apply agent-generated, risk tier labels
- Risk Scoring — calculate diff size, file impact, scope compliance

Standard CI Still Runs

• Linting • Tests • Build • Commit validation

THE 7 DEADLY SINS

Create product-v2 as new repo

Use SemVer tags in same repo

Push directly to main/dev

Always use PRs + protection rules

Tags without GitHub Releases

Create Release for every tag

Skip CHANGELOG

Update CHANGELOG with every release

Branches live forever

48-hour TTL, enforce cleanup

Squash merge to main

Use merge commit, preserve history

No AGENTS.md

Document agent rules explicitly

SCORE YOUR SETUP

Org metadata	0-10
Repo descriptions	0-10
Branch model	0-10
Architecture scaffold	0-10
AGENTS.md	0-10
PR templates	0-10
Commit conventions	0-10
Release protocol	0-10
Documentation	0-10
Teams & permissions	0-10

Final Score Ranges:

0-30: Beginner
30-60: Functional
60-80: Professional
80-100: AI-Native Production Grade

GET THE FULL GUIDE

- Score your own GitHub setup with the 14-dimension rubric
- Get the complete AGENTS.md template
- Access branch protection configurations
- Download release checklist and CI/CD examples

Download Link in Description

Subscribe for more AI-native systems architecture